

ASSIGNMENT

Ques 1 Overview of Hardware components : Introduction to mother-board, CPU, RAM.

→ Hardware components contains all physical parts of a computer system. This includes input, processor & output.

Hardware components :

- (1) CPU
- (2) Motherboard
- (3) RAM
- (4) Case Fan connector

* Introduction to Motherboard :

→ Motherboard is the central unit of computer that connect CPU, RAM & storage.

Main parts of ~~cpu~~ motherboard :

- CPU socket
- chip set
- RAM
- Case Fan connector
- PS/2 port
- SATA port.

* CPU : (central processing unit) :

→ It is a processor that acts as a brain of computer.

→ It contains PGA, LGA, BGA.

LGA - Land Grid Array, in electronics language, it is called FC (flip chip) - LGA

PGA - Pin Grid Array (PGA), in electronics languages, it is called FC - PGA

BGA - Ball pin Grid Array (BGA), in electronics language, it is called FC - BGA.

Component of CPU:

- cache
- ALU (Arithmetic logic & Unit)
- CU (control unit)
- It includes companies like AMD & Intel.
- AMD is used for normal use.
- Intel is used for rough use.

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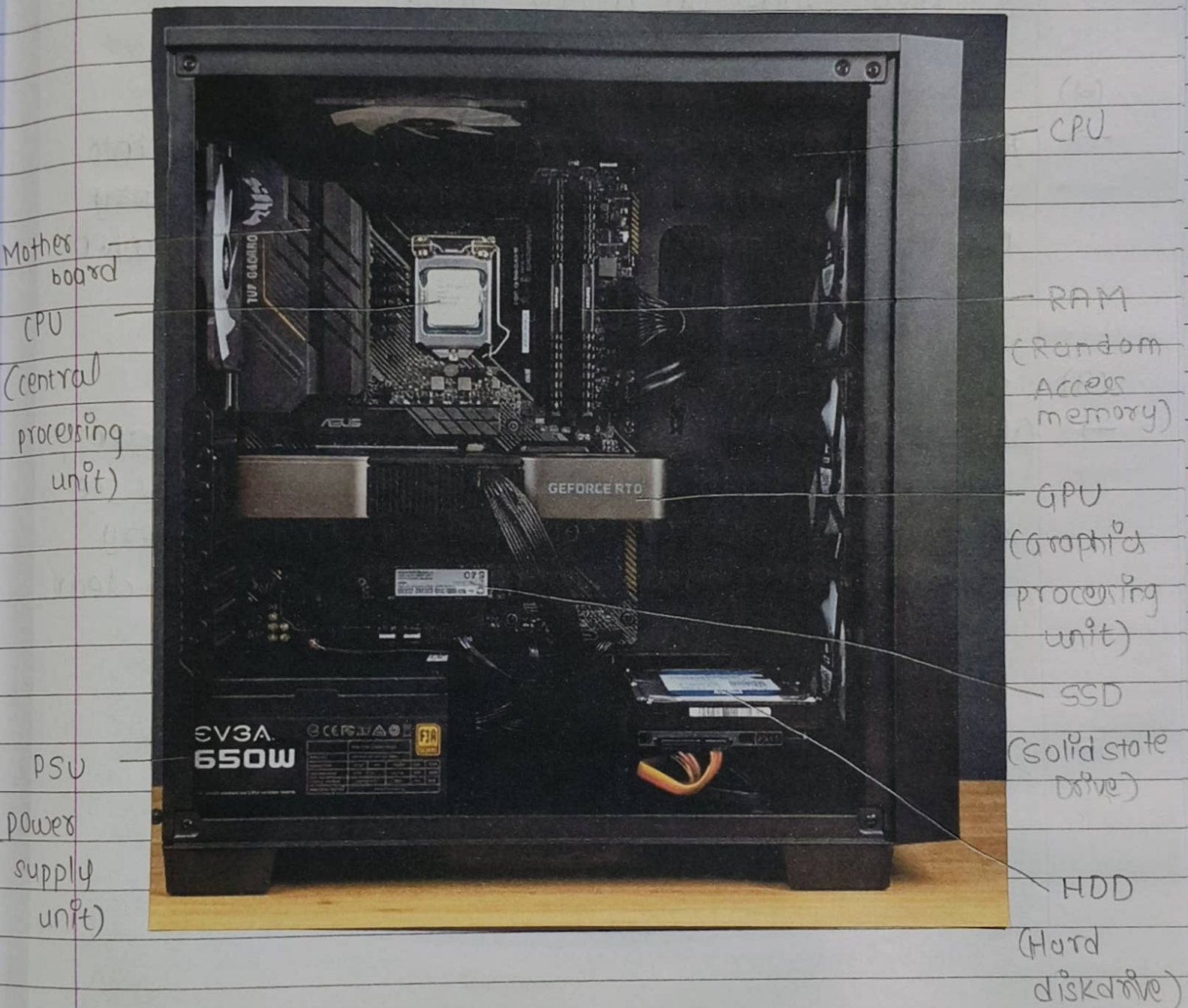
Components of ~~Hard~~ Motherboard:

- Integrated motherboard
- non-Integrated motherboard.
- chipset is a set of IC,
- The combination of the North & Southbridge in a computer is called chipset.

ASSIGNMENT CONTINUE

* Tasks:

Day 1: (1) Draw a labeled diagram of a desktop PC, clearly making the motherboard, CPU, RAM, GPU & both HDD & SSD storage components.



ASSIGNMENT DAY 1

Day 1(2) Create a comparison table showing the main difference between HDD & SSD storage in terms of speed, durability & typical use cases.

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower. Reads & writes data in about 80-160 MB/s. Files take longer to open.	Super Fast. Reads & writes data at 500-700 MB/s. Booting up takes seconds.
Durability	Fragile. Contains moving parts that can break if dropped or bumped while working.	Highly durable. Shock-resistant, with no moving parts.
Typical cases	Bulk storage like photos, movies.	Performance Device. For running operating system.

Day 1: (3) Write a short explanation describing the role of the CPU, RAM, & GPU in running a game like PUBG or Free Fire on a PC.

(1) CPU: (Central processing Unit)

→ CPU acts as the brain of the computer. It controls the game of rules, controls how enemies & players move, & keep track on match actions.

(2) RAM (Random Access Memory):

→ RAM is like the game graphics so it can grab into the split second. & having enough RAM ~~memory~~ keeps the game smooth without freezing when the game runs into the new area.

(3) GPU (Graphics processing unit):

→ GPU is the artist of computer. It controls 3D trees, building, explosions & characters that you see on your screen thousand of times, every minute, ^{to} make your game look sharp & clear.

Q1(a): Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading & explain why for each.

- Graphics card (GPU):

→ This controls how ~~the~~ game looks. upgrading it makes the game run smoothly without lag, keep the grass & players sharp, & stops the match from stuttering.

- Processor (CPU):

→ This acts as the brain of the PC. It handles the computer player (AI), referee decisions, & match rules. A faster CPU keeps the game flowing fast.

- Memory (RAM):

→ This lets your PC run heavy games without running out of memory. Upgrading upto atleast 16GB. & ~~stops~~ stops the game from freezing.

- SSD (Storage Drive):

→ This makes everything load faster. & loading screens disappear in seconds.